



Johanne Vincent

Assistant Professor in
Computer Science at
IMT ATLANTIQUE

- Date of Birth:** Jul. 1, 1983
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- Google Scholar:** Johanne Vincent
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Interests

- Computer Science
- Cybersecurity
- Privacy protection
- Graphic programming

Skills

Programming:

- C++ ● ● ● ● ●
- Python ● ● ● ● ●
- LaTeX ● ● ● ● ●
- Bash scripting ● ● ● ● ●

Tools:

- MATLAB ● ● ● ● ●
- GNU Radio ● ● ● ● ●
- Metasploit ● ● ● ● ●

Operating systems:

- Linux ● ● ● ● ●
- Windows ● ● ● ● ●
- Android ● ● ● ● ●

Working Experience

- Jul, 2015 – ongoing** **Assistant Professor** **IMT Atlantique, FR**
Assistant professor at IMT Atlantique in the Computer Science Department. Major research areas includes cybersecurity and privacy protection.
Member of the «sécurité et Résilience des Systèmes d'Information» (IRIS) team at the CNRS Lab-STICC, UMR 6285.
- Aug, 2014 – Jun, 2015** **Research Engineer** **SWID, FR**
R&D project manager on the DGA RAPID FRAG&TAG. Research and development activities around digital identity and the Internet of Things.
- Jun, 2013 – Jun, 2014** **Post-Doc Fellow** **Orange Labs, FR**
Title: «Identity management based on a Trusted Identity Module». Working within the Presence, Identity and Access (PIA) team.
- Jan, 2010 – Jun, 2013** **PhD candidate** **SFR, FR**
CIFRE doctoral student for SFR within the CNRS GREYC Laboratory, UMR 6072, at ENSICAEN (see subject below).
- Jan, 2007 – Dec, 2009** **Systems and Security Engineer** **Beijaflore Architecture des Systèmes d'Information (ASI), FR**
Worked on several missions including :
 - UNIX systems administrator for Voyages-sncf.com,
 - Windows Systems Administrator for AXA Group Solutions on distributed computing platforms for Risk Management,
 - Systems and security administrator for AXA Technology Services, operational security and authorizations for the AXA Group.

Education

Postgraduate Studies

- 2014** **Qualification as a lecturer** **27th CNU Section**
- 2010 – 2013** **Ph.D. in Computer Science** **SFR & GREYC Laboratory, FR**
Title: «Identité numérique en contexte télécom».
Supervisors: Prof. Christophe Rosenberger, Prof. Marc Pasquet, Mr. Kourosh Teimoorzadeh (SFR) and Dr. Christine Porquet.
Jury: Prof. Pacal Berthomé (president), Prof. Yves Deswartes, Prof. Maryline Laurent, Prof. Benjamin Nguyen

Security Digital Identity Privacy Telecommunications

- 2003 – 2006** **Engineering degree in Computer Science** **Ecole Nationale Supérieure d'Ingénieurs de Bourges (ENSIB) now known as INSA Centre Val de Loire**
Grade: Valedictorian in the Systems and Network Security major

System Security Network Security

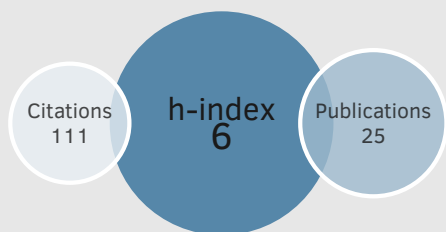
Undergraduate Study

- 2001 – 2003** **Classes Préparatoires aux Grandes Écoles (CPGE)** **A.R. Lesage Highschool, FR (56)**
Option: Mathematics, Physics, and Engineering Science. (*MPSI-MP*)
- 2001** **French Baccalauréat** **Charles de Gaulle Highschool, FR (56)**
Option: Sciences stream (*BAC S*).
Grade: high honors (*mention bien*).

Short Bio

Johanne is currently a lecturer at IMT Atlantique in the Computer Science department. Her main areas of interest are cybersecurity, and in particular privacy mechanisms and the protection of minorities.

Metrics



Profiles



Languages

French (Mother Tongue)

English (Cambridge English C1, 2006)

Personal

Johanne lives with her partner and their fabulously fluffy Akita Inu, Ôkami. When she's not busy decoding the secret language of tail wags and paw shakes, Johanne enjoys playing and learning to develop video games (when she has time), she also likes pop culture movies and TV shows (especially those that deals with feminist and queer representation). She has recently found herself a new hobby to dive in: self defense classes. Yes, you heard it right – the girl who used to bore unsuspecting victims with obscure airplane trivia can now kick butt and take names.

Teaching Experience

Postgraduate Courses

IMT Atlantique	Advanced C++ Programming (Coordinator)	Fall, Spring
	Introduction to the C++ programming language, STL, Modern C++, Strong focus on best software engineering practice, Design Patterns.	
	Advanced cryptology and data protection (Coordinator)	Spring
ENSICAEN	Introduction to Message Authentication Codes, Hash functions, Arithmetic, Homomorphic Encryption, Data Anonymization, Forensics.	
	Stochastic modeling and analysis	Fall'21&22
	Stochastic modelling, Random processes, Markov chains, Statistical quantities, Random signals, Adaptive filters, Practical implementation, General estimation theory.	
	Digital Identity	Fall'12&13
	Introduction to identity management, authentication, authorization, federation frameworks and SSO.	

Undergraduate Courses

IMT Atlantique	SIT 220: Information systems security (coordinator)	Spring
	Cybersecurity, cryptography, system security, network security.	
	MGP 210: Project Management (coordinator)	Fall
	Project management, planning, risk management, leading meetings.	
	SIT160: Cybersecurity awareness (coordinator)	Spring
	Introduction to cybersecurity concepts and awareness.	
	CODEVSI: Development project (steering committee)	Spring
	Project Management, technical specification, planing, writing technical reports and documentations.	
	SAR: Acoustics physics (coordinator)	Spring'2020, Spring'21
	Acoustic signal processing, modulation/demodulation, acoustic signal transmission, GNU Radio.	
	MTS110 Deterministic Signal Processing	Spring'2016 to Spring'21
	Signal, functions and distributions, time-frequency representation, analog/digital, filtering, Matlab/Octave.	

Academic Supervision

2024–ongoing	Towards a new generation of probes for monitoring encrypted traffic	Ph.D.
2022–ongoing	Authentication and control of inter-domains with guaranteed privacy in 5G networks.	Ph.D.
2022–ongoing	Ensuring privacy protection of people with disabilities when outsourcing data in IoT systems for ambient assisted living.	Ph.D.
2022–2025	Intrusion detection in cyber physical systems for enhanced cybersecurity of hospitals : a study of water management system.	Ph.D.
	Defended	

Invited Talks

May 2023	Vulgarization talk	Asso Labs BREST.
	Gave a two hours lecture on current privacy threats and challenges.	

References

Ref. 1	Prof. Thierry Duval thierry.duval@imt-atlantique.fr	IMT Atlantique, France
Ref. 1	Prof. Gouenou Coatrieux gouenou.coatrieux@imt-atlantique.fr	IMT Atlantique, France
Ref. 3	Prof. Caroline Fontaine caroline.fontaine@lsv.fr	LSV, CNRS & ENS Paris-Saclay, France

Publications

Journals

- Jacques Simonin, Pierre-Yves Pillain, Didier Guériot, and Johanne Vincent. Information system services generation of business services specification and based on a system-of-services logical architecture pattern. *International Journal of Cooperative Information Systems*, 29(03):2050002, 2020

International Conferences

- Henry Chima Ukwuoma, Gilles Dusserre, Gouenou Coatrieux, and Johanne Vincent. Digital twin based anomaly detection for an improved cyber security in a water distribution system. In *TIEMS 2023-The International Emergency Management Society Hybrid Annual Conference*, 2023
- Johanne Vincent, Wei Pan, and Gouenou Coatrieux. Privacy protection and security in ehealth cloud platform for medical image sharing. In *2016 2nd International Conference on Advanced Technologies for Signal and Image Processing (ATSIP)*, pages 93–96. IEEE, 2016
- Kent Are Varmedal, Henning Klevjer, Joakim Hovlandsvåg, Audun Jøsang, Johanne Vincent, and Laurent Miralabé. The offpad: Requirements and usage. In *Network and System Security: 7th International Conference, NSS 2013, Madrid, Spain, June 3-4, 2013. Proceedings 7*, pages 80–93. Springer Berlin Heidelberg, 2013
- Johanne Vincent, Vincent Alimi, Aude Plateaux, Chrystel Gaber, and Marc Pasquet. A mobile payment evaluation based on a digital identity representation. In *2012 International Conference on Collaboration Technologies and Systems (CTS)*, pages 410–418. IEEE, 2012
- Johanne Vincent, Christine Porquet, Maroua Borsali, and Harold Leboulanger. Privacy protection for smart-phones: an ontology-based firewall. In *Information Security Theory and Practice. Security and Privacy of Mobile Devices in Wireless Communication: 5th IFIP WG 11.2 International Workshop, WISTP 2011, Heraklion, Crete, Greece, June 1-3, 2011. Proceedings 5*, pages 371–380. Springer Berlin Heidelberg, 2011
- Johanne Vincent, Christine Porquet, and Idriss Oulmakhzoune. Ontology-based privacy protection for smart-phone: A firewall implementation. In *International Conference on Secure Networking and Applications (ICSNA)*, pages 3–pages, 2011
- Johanne Vincent, Marc Pasquet, and W Chaisantikulwat. Security and privacy analysis of a physical access control solution. In *IEEE International Conference on Information Theory and Information Security (ICITIS)*, 2011
- Jean Claude Paillès and Johanne Vincent. A privacy-enhanced m-transactions architecture for awareness and trust. In *2010 Third International Conference on Advances in Human-Oriented and Personalized Mechanisms, Technologies and Services*, pages 93–98. IEEE, 2010
- Johanne Vincent, Jean-Philippe Wary, and Marc Pasquet. Cartography of actors and roles in identity management solutions. In *Future Internet Symposium*, 2010

National Conferences

- Henry Chima Ukwuoma, Gilles Dusserre, Gouenou Coatrieux, and Johanne Vincent. Digital twin and intrusion detection in a cyber physical system. In *Colloque IMT 2023:«Sécurité et Résilience»*, 2023
- Henry Chima Ukwuoma, Gilles Dusserre, Gouenou Coatrieux, and Johanne Vincent. Digital twin and intrusion detection in a water distribution system. In *Séminaire IMT «Data Analytics & Artificial Intelligence»*, 2023
- J Vossaert, P Verhaeghe, B Decker, and V Naessens. User-centric identity management using trusted identity module-binding mobile phone secure elements to the openid connect protocol. In *SARSSI 2014, 9th Conference on Network Security Architectures and Information Systems*
- Johanne Vincent, Tom Dubin, and Christine Porquet. Protection de la vie privée basée sur des ontologies dans un système android. In *APVP 2012 (Atelier Protection de la Vie Privée, 3eme édition)*, pages 4–pages, 2012
- Johanne Vincent, Kourosh Teimoorzadeh, Christophe Rosenberger, and Marc Pasquet. Analyse de la sécurité et de la protection de la vie privée des systèmes de gestion d'identités. In *7ème Conférence sur la Sécurité des Architectures Réseaux et Systèmes d'Information (SAR SSI)*, 2012

Thesis

- Johanne Vincent. *Identité numérique en contexte Télécom*. PhD thesis, Université de Caen, 2013

Patent

- Kurosh Teimoorzadeh, Johanne Vincent, Jean-Philippe Wary, Jean-Claude Paillès, and Christophe Rosenberger. System for managing digital identity, 2013